



NEET(UG)
MAJOR

This Booklet contains 20 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and this Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
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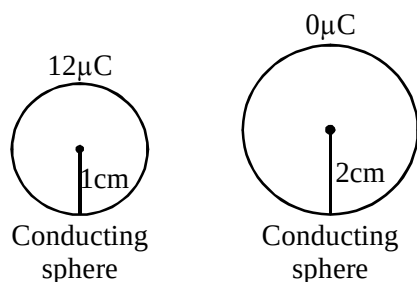
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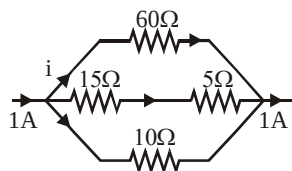
LEADER & ACHIEVER COURSE
(PHASE : MLI,J,K,M,N,O,R,S,MAZG,H,I,J,K,L,M,T,U,M4AA2A,M2AP1A,M2AP1B)

ALLEN MAJOR TEST		DATE : 03 - 03 - 2020	
SYLLABUS – 04			
PHYSICS : Electrostatics, Capacitor, Current electricity and Heating Effects of Current.			
CHEMISTRY : Solid State, Chemical kinetics, Solutions, Electrochemistry, Surface chemistry			
BIOLOGY : Reproduction : (i) Reproduction in Organisms (ii) Sexual Reproduction in Flowering Plants (iii) Human Reproduction (iv) Reproductive Health (v) Evolution			

1. When these two spheres are made to touch with each other, then final charge on bigger sphere will be :



- (1) $0\mu\text{C}$ (2) $12\mu\text{C}$ (3) $4\mu\text{C}$ (4) $8\mu\text{C}$
2. An electron is released with a velocity of $5 \times 10^6 \text{ ms}^{-1}$ in an electric field of 10^3 NC^{-1} which has been applied so that its oppose motion. How much time could it take before it is brought to rest:-
- (1) $2.8 \times 10^{-8} \text{ s}$ (2) $1.8 \times 10^{-8} \text{ s}$
 (3) $6.4 \times 10^{-8} \text{ s}$ (4) $4.2 \times 10^{-8} \text{ s}$
3. Current flows through a metallic conductor whose area of grass-section, increases in the direction of the current, if we move in this direction :-
- (1) the current will change
 (2) the carrier density will increase
 (3) the drift velocity will remains same
 (4) the drift velocity will decrease
4. The magnitude of i is :-

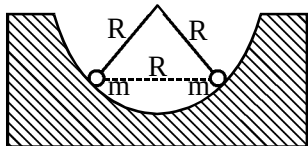


- (1) 0.1A (2) 0.3A
 (3) 0.6A (4) None of these
5. Drift speed of electrons is of the order of :-
- (1) Zero (2) 10^{-2} cm/sec
 (3) 10^5 m/sec (4) 10^0 m/sec
6. Three charges $4q$, Q and q are in a straight line in the position of 0 , $\ell/2$ and ℓ respectively. The resultant force on q will be zero, if $Q =$
- (1) $-q$ (2) $-2q$ (3) $-\frac{q}{2}$ (4) $4q$

7. An electric dipole, consisting of point charges q and $-q$ kept at a separation 'd' is placed in a uniform electric field with its axis inclined to the field direction at angle $\theta > 90^\circ$, but $\theta \neq 180^\circ$. The dipole experiences torque and rotates. Which of the following is/are true ?

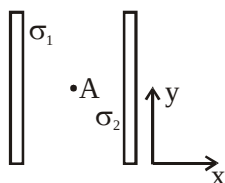
- (1) Angular momentum remains conserved
 (2) Angular momentum is not conserved.
 (3) Angular momentum is not conserved for rotation from $\theta > 90^\circ$ to $\theta = 90^\circ$, but it is conserved for rotation from $\theta = 90^\circ$ to $\theta < 90^\circ$.
 (4) Angular momentum is conserved for rotation from $\theta > 90^\circ$ to $\theta = 90^\circ$ but it is not conserved for rotation from $\theta = 90^\circ$ to $\theta = 0^\circ$.
8. Path of electron inside conductor between two consecutive collision when no potential difference exists between the ends of conductor :-
- (1) Stright line
 (2) Parabolic
 (3) Circular
 (4) Zig-zag
9. A current 2 A flows through a 2Ω resistor when connected across a battery. The same battery supplied a current 0.5 A when connected across a 9Ω resistor. The internal resistance of the battery is
- (1) 1Ω
 (2) 0.5Ω
 (3) $1/3 \Omega$
 (4) $1/4 \Omega$
10. In a Wheatstone's bridge all the four arms have equal resistance R . If the resistance of the galvanometer arm is also R , the equivalent resistance of the combination as seen by the battery is
- (1) $\frac{R}{2}$
 (2) R
 (3) $2 R$
 (4) $\frac{R}{4}$

11. Two identical small bead each have a mass m and charge q . When placed in a hemispherical bowl of Radius R . With frictionless, non conductive walls. The beads move and comes at equilibrium. The line joining the balls is horizontal and the distance between them is R . Neglect any induced charge on the hemispherical bowl. Then the charge on each bead is :



- (1) $q = R \left(\frac{mg}{k\sqrt{3}} \right)^{1/2}$ (2) $q = \left(R \frac{mg}{k\sqrt{3}} \right)^{1/2}$
 (3) $q = R \left(\frac{\sqrt{3}mg}{k} \right)^{1/2}$ (4) $q = \left(R \frac{\sqrt{3}mg}{k} \right)^{1/2}$

12. Two large conducting sheets are kept parallel to each other as shown. In equilibrium, the charge density on facing surfaces is σ_1 and σ_2 . Which of the following may not be electric field at A :-



- (1) $\frac{\sigma_1 \hat{i}}{\epsilon_0}$ (2) $\frac{-\sigma_2 \hat{i}}{\epsilon_0}$
 (3) $\frac{(\sigma_1 + \sigma_2) \hat{i}}{2\epsilon_0}$ (4) $\frac{(\sigma_1 - \sigma_2) \hat{i}}{2\epsilon_0}$

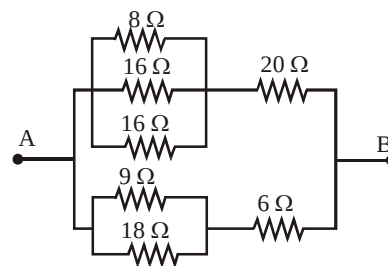
13. The resistance of a 20 cm, long wire is 5Ω . It is stretched upto 40 cm. The resistance of wire will now be :- (in ohm)

- (1) 5 (2) 10 (3) 20 (4) 200

14. A galvanometer has 36Ω resistance. If a 4Ω shunt is added to this, the fraction of current that passes through the shunt is :-

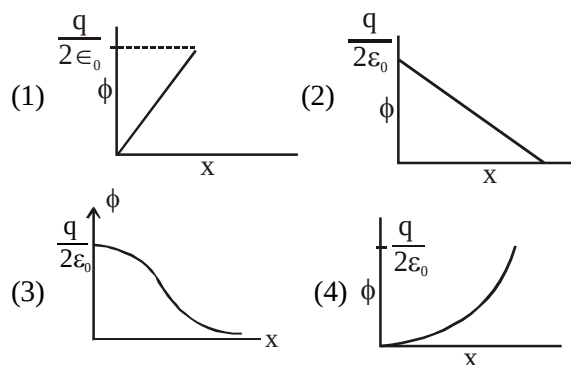
- (1) $\frac{1}{4}$ (2) $\frac{1}{9}$ (3) $\frac{1}{10}$ (4) $\frac{9}{10}$

15. The equivalent resistance of the arrangement of resistances shown in adjoining figure between the points A and B is

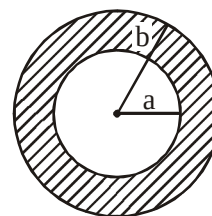


- (1) 6 ohm (2) 8 ohm (3) 16 ohm (4) 24 ohm

16. A point charge is placed at centre of a disc now it is moved away from centre along axis then graph between flux related to the disc and distance (x) from centre of disc is :

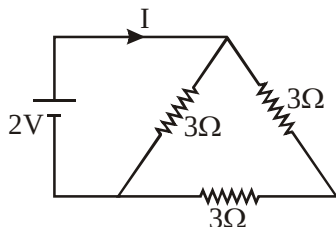


17. A spherical shell with an inner radius ' a ' and an outer radius ' b ' is made of conducting material. A point charge $+Q$ is placed at the centre of the spherical shell and a total charge $-q$ is placed on the shell. Charge $-q$ is distributed on the surfaces as

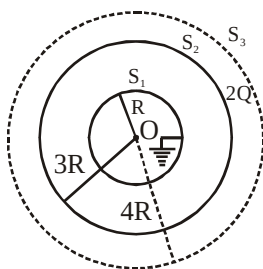


- (1) $-Q$ on inner surface, $-q$ on outer surface
 (2) $-Q$ on inner surface, $-q + Q$ on outer surface
 (3) $+Q$ on the inner surface, $-q - Q$ on the outer surface
 (4) The charge $-q$ is spread uniformly between inner and outer surface.

18. For a wire having constant resistance the graph between V^2 and I^2 is :-
 (1) Hyperbola (2) Rectangular Hyperbola
 (3) Ellipse (4) Straight line
19. What will be the power of a 100 W, 220 V lamp when used on 110 V power supply ?
 (1) 100 W (2) 75 W (3) 50 W (4) 25 W
20. The current in the following circuit is :-

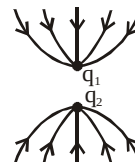


- (1) $\frac{1}{8}$ A (2) $\frac{2}{9}$ A (3) $\frac{2}{3}$ A (4) 1 A
21. In given diagram two concentric conducting spherical shells S_1 & S_2 are shown charge on S_2 is $2Q$ where S_1 is earthed find electric flux related to surface S_3

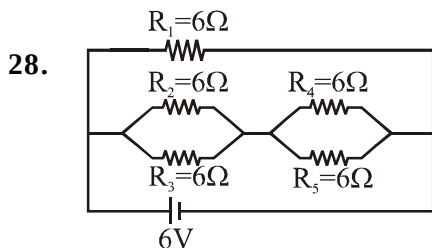


- (1) $\frac{1}{3} \frac{Q}{\epsilon_0}$ (2) $\frac{4}{3} \frac{Q}{\epsilon_0}$
 (3) $\frac{2Q}{\epsilon_0}$ (4) zero
22. A glass slab is put with in the plates of a charged parallel plates condenser. (not completed to battery) Which of the following quantities does not change.
 (1) energy of the condenser
 (2) capacity
 (3) intensity of electric field
 (4) charge

23. Resistance of a semiconductor, with increase in temperature :-
 (1) Increases
 (2) Remains constant
 (3) Decreases
 (4) Vary in complex way
24. In the adjoining figure, the electric field lines for charges q_1 and q_2 are shown. Identify the sign of the charges:



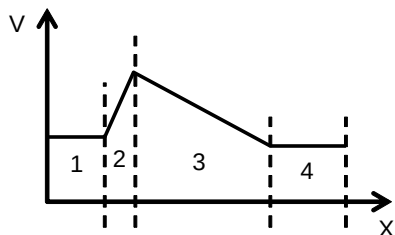
- (1) both negative
 (2) upper charge is negative and lower is positive
 (3) both positive
 (4) upper charge is positive and lower is negative
25. A battery has an emf 15 volt and internal resistance 1Ω . Is the terminal to terminal potential difference less than, equal to or greater than 15 volt if the current in the battery is:
 (i) from (-)ve to (+)ve terminal
 (ii) from (+)ve to (-)ve terminal
 (iii) zero current
 (1) less, greater, equal
 (2) less, less, equal
 (3) gerater, greater, equal
 (4) greater, less, equal
26. As one moves towards the centre from surface of a uniformly charged metallic sphere the potential-
 (1) Increase (2) Decrease
 (3) Remains constant (4) Zero
27. The work done in placing a charge of 8×10^{-18} coulomb on a condenser of capacity 100 micro-farad is :
 (1) 16×10^{-32} Joule
 (2) 3.1×10^{-26} Joule
 (3) 4×10^{-10} Joule
 (4) 32×10^{-32} Joule



Consider an electric circuit shown in figure, what is the potential difference across the resistor R_4 :

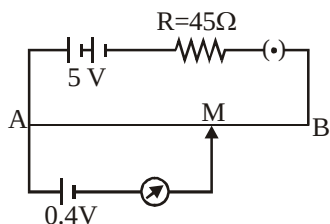
- (1) 6V (2) 4.5 V (3) 3V (4) 1.5 V

29. In which region magnitude of x-component of electric field is maximum, if potential (V) versus distance (X), graph is as shown?



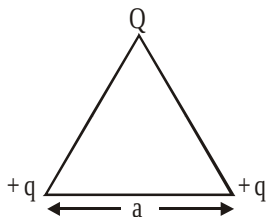
- (1) 1 (2) 2 (3) 3 (4) 4

30. In given figure, the potentiometer wire AB has a resistance of $5\ \Omega$ and length 10 m. The balancing length AM for the emf of 0.4 V is :-



- (1) 0.4 m (2) 4 m (3) 0.8 m (4) 8 m

31. Three charges Q , $+q$, and $+q$ are placed at the vertices of an equilateral triangle as shown in figure. If net electrostatic energy of this configuration is zero, then Q is equal to -



- (1) $-2q$ (2) $-\frac{q}{2}$ (3) $\frac{\sqrt{3}q}{2}$ (4) $-\frac{\sqrt{3}q}{2}$

32. Two spherical conductors each of capacity C are charged to potential V and $-V$. These are then connected by means of a fine wire. The loss of energy will be :-

- (1) zero (2) $\frac{1}{2}CV^2$ (3) CV^2 (4) $2CV^2$

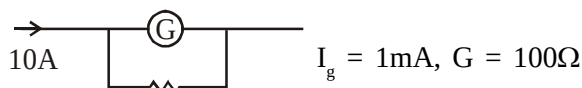
33. Kirchhoff's first law i.e., $\sum i = 0$ at a junction is based on the law of conservation of :-

- (1) Charge (2) Energy
(3) Momentum (4) Angular momentum

34. When an alpha-particle is accelerated by a PD of 3 volt, its energy is:

- (1) 1 eV (2) 5 eV (3) 3 eV (4) 6 eV

35. When the current through ammeter is giving full scale deflection then heat generated by shunt per unit time is :-



- (1) 1W
(2) 0.9999 W
(3) 1.0001 W
(4) no heat will be dissipated

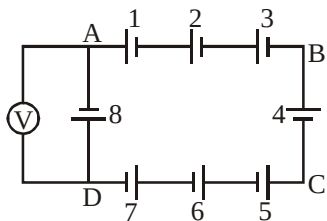
36. Two fixed charges $-2Q$ and Q are located at the points with coordinates $(-3a, 0)$ and $(3a, 0)$ in the x - y plane. Find the locus of all points in the x - y plane where the electric potential due to the charges is zero :-

- (1) $x^2 + y^2 - 10ax + 9a^2 = 0$
(2) $x^2 + 2y^2 - 10ax + 9a^2 = 0$
(3) $2x^2 + y^2 - 10ax + 9a^2 = 0$
(4) $x^2 + y^2 - 20ax + 9a^2 = 0$

37. There are two identical capacitors, the first one is uncharged and filled with a dielectric constant K while the other one is charged to potential V having air between its plates. If two capacitors are joined end to end, the common potential will be

- (1) $\frac{V}{K-1}$ (2) $\frac{KV}{V+1}$ (3) $\frac{KV}{K-1}$ (4) $\frac{V}{K+1}$

38. Eight cell marked 1 to 8, each of emf 5 V and internal resistance 0.2Ω are connected as shown. What is the reading of ideal voltmeter ?

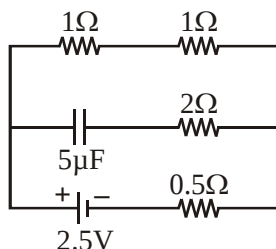


- (1) 40 V (2) 20 V (3) 5 V (4) Zero

39. A small electric dipole is of dipole moment p . The electric potential at a distance ' r ' from its centre and making an angle θ from the axis of dipole will be :-

- (1) $\frac{kp \sin \theta}{r^2}$
 (2) $\frac{kp \cos \theta}{r^2}$
 (3) $\frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$
 (4) $\frac{kp}{r^3} \sqrt{1 + 3 \sin^2 \theta}$

40. Find the final stored energy in the capacitor :-

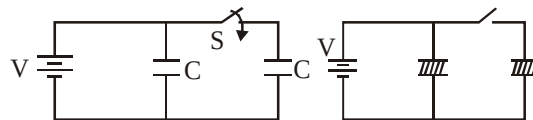


- (1) 10 μJ (2) 20 μJ (3) 30 μJ (4) 40 μJ

41. An electron enters in a uniform electric field maintained by parallel plates and of value $E \text{ Vm}^{-1}$ with a velocity $v \text{ ms}^{-1}$. If the plates are separated by a distance d metre, then magnitude of acceleration of the electron in the field is :-

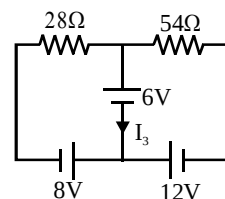
- (1) $\frac{Ee}{m}$ (2) $\frac{-Eev}{m}$ (3) $\frac{Ee}{md}$ (4) $Ee \frac{d}{m}$

42. Figure below shown two identical parallel plate capacitors connected to battery with switch S closed. The switch is now opened and the free space between the plates of capacitors is filled with a dielectric of dielectric constant 3. What will be the ratio of total electrostatic energy stored in both capacitors before and after the introduction of the dielectric :-



- (1) $\frac{3}{5}$ (2) $\frac{2}{5}$ (3) $\frac{1}{5}$ (4) $\frac{4}{5}$

43. Consider the circuit shown in the figure. The current I_3 is equal to :-

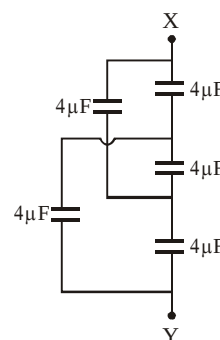


- (1) 5 amp (2) 3 amp
 (3) -3 amp (4) -5/6 amp

44. A spherical charged conductor has surface charge density σ . The electric field on its surface is E and electric potential of conductor is V . Now the radius of the sphere is halved keeping the charge to be constant. The new values of electric field and potential would be :

- (1) $2E, 2V$ (2) $4E, 2V$ (3) $4E, 4V$ (4) $2E, 4V$

45. Equivalent capacitance between X and Y points in the given figure is :-



- (1) $2 \mu F$ (2) $4 \mu F$ (3) $3 \mu F$ (4) $1 \mu F$

46. Which statement is not correct ?
 (1) Conductance of an electrolytic solution increases with dilution.
 (2) Conductance of an electrolytic solution decreases with dilution
 (3) Specific conductance of an electrolytic solution decreases with dilution
 (4) Equivalent conductance of an electrolytic solution increases with dilution
47. Which of the following salts has same value of Van't Hoff's factor as that of $K_3[Fe(CN)_6]$?
 (1) Na_2SO_4 (2) $Al(NO_3)_3$
 (3) $Al_2(SO_4)_3$ (4) $NaCl$
48. How much charge is required for the oxidation of 1 mole of $S_2O_3^{2-}$ to $S_4O_6^{2-}$?
 (1) 96500 C (2) 48250 C
 (3) 386000 C (4) 193000 C
49. The edge length of BCC unit cell is 508 pm then the approx radius of atom is :-
 (1) 254 pm (2) 398 pm
 (3) 220 pm (4) 180 pm
50. Which of the following acts as an emulsifier ?
 (1) Soap (2) Water
 (3) Oil (4) $NaCl$
51. $A \longrightarrow B$
 If $K = 0.5 \times 10^{-3} \text{ M sec}^{-1}$ then concentration of B after 10 minute is :-
 (1) 0.3 M (2) $0.5 \times 10^{-2} \text{ M}$
 (3) $0.5 \times 10^{-3} \text{ M}$ (4) 0.5 M
52. If the total vapour pressure of liquid mixture A and B is given by the equation $P = 180 x_A + 90$ then the ratio of the vapour pressure of the pure liquids A and B is given by :-
 (1) 3 : 2 (2) 4 : 1 (3) 3 : 1 (4) 6 : 2
53. During the electrolysis of water, oxygen is liberated at anode :
 $2H_2O_{(l)} \longrightarrow 2H_{2(g)} + O_{2(g)}$
 Calculate the volume of $O_{2(g)}$ produced at STP when a current of 30 A is passed through acidulated water for 2 hours?
 (1) 12.544 L (2) 3.136 L
 (3) 6.272 L (4) 15.48 L
54. In ionic crystal AB if radius of cation and anion is 30 pm and 60 pm respectively then crystal structure is :-
 (1) ZnS type (2) $NaCl$ type
 (3) $CsCl$ type (4) B_2O_3 type
55. For a colloids $[Fe(OH)_3]Fe^{3+} | Cl^-$, which of the following electrolytes will have maximum coagulating value ?
 (1) Na_2S (2) Na_3PO_4
 (3) $K_4[Fe(CN)_6]$ (4) $NaCl$
56. How many moles of Pt may be deposited at cathode when 0.8 F charge is passed through 1M solution of Pt^{+4} ions?
 (1) 1 mole (2) 0.2 mole
 (3) 0.4 mole (4) 0.8 mole
57. Density of sulphuric acid solution is 1.2 g/mL and it is 40% H_2SO_4 by weight. Determine molarity of this solution.
 (1) 4.9 M (2) 9.8 M
 (3) 2.45 M (4) 0.1 M
58. Correct order of reducing power of the following metals is :-
 Al Cu Fe Mg Zn
 (1) $Mg > Al > Zn > Fe > Cu$
 (2) $Mg > Al > Zn > Cu > Fe$
 (3) $Mg < Al < Zn < Fe < Cu$
 (4) $Zn > Al > Mg > Fe > Cu$
59. The number of ZnS formula units present in a unit cell of ZnS are :-
 (1) 2 (2) 1 (3) 4 (4) 8
60. For a given reaction, energy of activation for forward reaction (E_{af}) is 80 $KJ \text{ mol}^{-1}$ and $\Delta H = -40 \text{ KJ mol}^{-1}$ for the reaction. A catalyst lowers (E_{af}) by 20 $KJ \text{ mol}^{-1}$. The ratio of energy of activation for reverse reaction before and after addition of catalyst is :-
 (1) 1.0 (2) 0.5
 (3) 1.2 (4) 2.0

61. The half cell reactions are :
 (i) $\text{Fe}^{+2}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Fe}(\text{s}); \quad E^\circ = -0.44 \text{ V}$
 (ii) $2\text{H}^+(\text{aq}) + \frac{1}{2}\text{O}_2(\text{g}) + 2\text{e}^- \longrightarrow \text{H}_2\text{O}(\ell); \quad E^\circ = 1.23 \text{ V}$
 The E° for the reaction
 $\text{Fe}(\text{s}) + 2\text{H}^+(\text{aq}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{Fe}^{+2}(\text{aq}) + \text{H}_2\text{O}(\ell)$
 is :-
 (1) 1.67 V (2) -1.67 V
 (3) 0.79 V (4) -0.79 V
62. The molecular weight of NaCl, determined by studying freezing point depression of its 0.5% aqueous solution is 30. The apparent degree of dissociation of NaCl is :-
 (1) 0.95 (2) 0.50 (3) 0.60 (4) 0.30
63. The emf (E°) of the following cells are :
 $\text{Ag} | \text{Ag}^+(1\text{M}) || \text{Cu}^{+2}(1\text{M}) | \text{Cu}; \quad E^\circ = -0.46 \text{ V}$
 $\text{Zn} | \text{Zn}^{+2}(1\text{M}) || \text{Cu}^{+2}(1\text{M}) | \text{Cu}; \quad E^\circ = 1.10 \text{ V}$
 Calculate emf of the cell
 $\text{Zn} | \text{Zn}^{+2}(1\text{M}) || \text{Ag}^+(1\text{M}) | \text{Ag}$
 (1) 0.92 V (2) 0.32 V
 (3) 1.56 V (4) 0.72 V
64. In a cubical unit cell, A atoms occupy 50% of corners, while B atoms at all the face centres and C at 50% of edge centres then the formula of solid is:-
 (1) AB_6C_3 (2) $\text{A}_2\text{B}_3\text{C}_3$
 (3) AB_6C_6 (4) A_6BC
65. The rate law for a reaction $\text{A} + \text{B} \rightarrow \text{product}$ is $r = K [\text{A}]^2[\text{B}]^{-1}$ then which of the following statement is false :-
 (1) If [B] is held constant while [A] is doubled, the rate of reaction will be quadrupled
 (2) If [A] is held constant while [B] is reduced to one quarter, the rate will be quadrupled
 (3) If [A] & [B] are both doubled, the rate of reaction will be quadrupled
 (4) This is a first order reaction
66. Osmotic pressure of a solution containing 5g of protein in 500 ml of solution at 300 K is 1.66 bar. This protein contain 48% carbon. What is number of carbon atoms present in one molecule of protein:-
 (1) 2 (2) 4 (3) 6 (4) 8

67. Given : $\Lambda_{\text{M}(\text{Mg}^{+2})}^\infty = 106.1 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$,
 $\Lambda_{\text{M}(\text{SO}_4^{+2})}^\infty = 160 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$,
 $\Lambda_{\text{M}(\text{Al}^{+3})}^\infty = 189 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$.
 Calculate $\Lambda_{\text{M}(\text{Al}_2(\text{SO}_4)_3)}^\infty$ and $\Lambda_{\text{M}(\text{MgSO}_4)}^\infty$ in $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$:-
 (1) 858, 266.1 (2) 133.05, 429
 (3) 429, 266.1 (4) 858, 133.05
68. Total volume of atoms present in a HCP unit cell of a metal is :-
 (1) $\frac{20}{3} \pi r^3$ (2) $\frac{16}{3} \pi r^3$ (3) $\frac{24}{3} \pi r^3$ (4) $\frac{12}{3} \pi r^3$
69. During adsorption :-
 (1) $T\Delta S$ is positive
 (2) $(\Delta H - T\Delta S)$ is negative
 (3) ΔH is positive
 (4) $T\Delta S$ and ΔH is zero
70. Which of the following represents Wilhelmy equation (where $[\text{A}]_t$ & $[\text{A}]_0$ have usual meaning):-
 (1) $[\text{A}]_t = [\text{A}]_0 e^{-Kt}$ (2) $[\text{A}]_t = [\text{A}]_0 e^{Kt}$
 (3) $\frac{1}{[\text{A}]_t} - \frac{1}{[\text{A}]_0} = Kt$ (4) $[\text{A}]_t = [\text{A}]_0 + K \cdot t$
71. Which is correctly matched for homogeneous solution?
- | | | |
|-----|--|--|
| (1) | $\text{HNO}_3 + \text{H}_2\text{O}$ | $\Delta H = +ve$
$\Delta S = +ve$
$\Delta G = -ve$ |
| (2) | $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$ | $\Delta H = +ve$
$\Delta S = +ve$
$\Delta G = +ve$ at high temperature |
| (3) | $\text{CHCl}_3 + \text{Acetone}$ | $\Delta H = -ve$
$\Delta S = +ve$
$\Delta G = -ve$ |
| (4) | Toluene + Benzene | $\Delta H = +ve$
$\Delta S = +ve$
$\Delta G = -ve$ |
72. The resistance of conductivity cell when filled with 0.01 M solution of an electrolyte A_2B is 150Ω at 20°C . The same conductivity cell when filled 0.02 M XY_2 solution has resistance of 600Ω . The specific conductivity of 0.01 M A_2B solution is 0.002 S cm^{-1} . Calculate the Λ_{M} of 0.02 M XY_2 solution :-
 (1) $12.5 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $25 \text{ S cm}^2 \text{ mol}^{-1}$
 (3) $20 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $50 \text{ S cm}^2 \text{ mol}^{-1}$

73. If KCl is doped with 10^{-5} mol % of CdCl_2 , the concentration of cation vacancies will be :-
 (1) $6.023 \times 10^{15} \text{ mol}^{-1}$ (2) $6.023 \times 10^{17} \text{ mol}^{-1}$
 (3) $6.023 \times 10^{16} \text{ mol}^{-1}$ (4) $6.023 \times 10^{18} \text{ mol}^{-1}$
74. Activated charcoal is used to remove colouring matter from pure substance. It works by the principle of:-
 (1) Oxidation (2) Reduction
 (3) Bleaching (4) Adsorption
75. For a reaction $E_a = 0$ and $A = 3.2 \times 10^4 \text{ s}^{-1}$ at 300K. Then value of K at 310K would be :-
 (1) $6.4 \times 10^4 \text{ s}^{-1}$ (2) $3.2 \times 10^4 \text{ s}^{-1}$
 (3) $3.2 \times 10^8 \text{ s}^{-1}$ (4) $3.2 \times 10^5 \text{ s}^{-1}$
76. Increasing the temperature of an aqueous solution will cause :-
 (1) Decrease in molality
 (2) Decrease in molarity
 (3) Decrease in mole fraction
 (4) Decrease in % (w/w)
77. The measured voltage of the electrochemical cell
 $\text{Cu(s)} + 2\text{Ag}^+ (1\text{M}) \rightleftharpoons 2\text{Ag(s)} + \text{Cu}^{2+} (1\text{M})$
 is 0.54 V at 25°C. Calculate the standard free energy change of the reaction :
 (1) -104220 J (2) -52110 J
 (3) -24840 J (4) -208440 J
78. In Schottky defect :-
 (1) Density remains unchanged
 (2) Density gets decreased
 (3) Density gets increased
 (4) Limiting radius ratio is low
79. At high pressure, Langmuir adsorption isotherm takes the form :-
 (1) $\frac{x}{m} = \frac{ap}{1+bp}$ (2) $\frac{x}{m} = \frac{a}{b}$
 (3) $\frac{x}{m} = ap$ (4) $\frac{m}{x} = \frac{b}{a} + \frac{1}{ap}$
80. Unit of rate constant of elementary reaction depends upon the :-
 (1) Temperature of the reaction
 (2) Concentration of reactant
 (3) Activation energy of the reaction
 (4) Molecularity of the reaction
81. 0.1 M NaCl and 0.1 M glucose would exert :-
 (1) Same osmotic pressure
 (2) Osmotic pressure in the ratio 1 : 3
 (3) Osmotic pressure in the ratio 2 : 1
 (4) Osmotic pressure in the ratio 1 : 2
82. The standard voltage of the cell :-
 $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\ell)$
 is 1.18 V at 27°C. Calculate the equilibrium constant of the reaction :-
 (1) 1×10^{80} (2) 7.49×10^{44}
 (3) 4.56×10^{71} (4) 6.4×10^{66}
83. Frenkel defect is generally observed in :-
 (1) NaCl (2) CsCl (3) ZnS (4) KCl
84. Which of the following substances gives a positively charged sol ?
 (1) Gold
 (2) A metal sulphite
 (3) Ferric hydroxide
 (4) An acidic dye in basic medium
85. $\text{N}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{NO}_{(\text{g})}$
 If $\frac{d[\text{NO}]}{dt} = 2 \times 10^{-4} \text{ M sec}^{-1}$ then $-\frac{d[\text{N}_2]}{dt}$ is :-
 (1) $4 \times 10^{-4} \text{ M sec}^{-1}$ (2) $2 \times 10^{-4} \text{ M sec}^{-1}$
 (3) $1 \times 10^{-4} \text{ M sec}^{-1}$ (4) $3 \times 10^{-4} \text{ M sec}^{-1}$
86. When 30 g of a non-volatile solute is dissolved in 250 g of cyclohexane, the freezing point of the latter drops from 6.5°C to 2.5°C. What is the molecular weight of solution ? (K_f for cyclohexane = 20)
 (1) 150 (2) 500 (3) 24 (4) 600
87. $\text{Cu}^+ + \text{e}^- \longrightarrow \text{Cu}$ $E_1^\circ = x_1 \text{ V}$
 $\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$ $E_2^\circ = x_2 \text{ V}$ then for
 $\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$ E° in (volts) will be :-
 (1) $x_1 - 2x_2$ (2) $x_1 + 2x_2$
 (3) $x_1 - x_2$ (4) $2x_2 - x_1$
88. AB crystal has CsCl type structure. If edge length of unit cell is 100 pm then nearest distance between cation and anion is :-
 (1) 50 pm (2) 100 pm
 (3) 173.2 pm (4) 86.6 pm
89. Light scattered by colloidal particles is :-
 (1) Visible to naked eye
 (2) Not visible by any medium
 (3) Visible under ordinary microscope
 (4) Visible under ultra-microscope
90. Ratio of $t_{7/8}$ & $t_{1/2}$ for a first order reaction is :-
 (1) 3 : 1 (2) 8 : 1 (3) 7 : 4 (4) 4 : 7

- 91.** Choose the incorrect statement :-
 (1) Sertoli cells line seminiferous tubule and provide nutrition to the germ cells.
 (2) Regions inside the seminiferous tubules called interstitial spaces, contain interstitial cells.
 (3) Leydig cells synthesise and secrete testicular hormones, androgens.
 (4) Immunologically competent cells are also present in interstitial spaces.
- 92.** Which of the following is non-steroidal contraceptive pill?
 (1) Saheli (2) IUDs
 (3) Vaults (4) Diaphragms
- 93.** First haploid structure produced during spermatogenesis is called as :-
 (1) Spermatid
 (2) Spermatozoa
 (3) Primary spermatocyte
 (4) Secondary spermatocyte
- 94.** Which of the following is a method of *in-vitro* fertilization ?
 (1) Gamete intra fallopian transfer (GIFT)
 (2) Intra uterine insemination (IUI)
 (3) Amniocentesis
 (4) Intra cytoplasmic sperm injection (ICSI)
- 95.** The study of history of life forms on earth, is called as :-
 (1) Demography
 (2) Developmental Biology
 (3) Evolutionary Biology
 (4) Pedigree
- 96.** Read the statements carefully from A-D and choose the option which include correct statements :-
 (A) The skull of baby chimpanzee is more like adult chimpanzee skull than adult human skull.
 (B) The first mammals were like **shrews**.
 (C) When reptiles came down, mammals took over this earth.
 (D) Darwinian variations were small and directionless.
 (1) A and C (2) B and D
 (3) B and C (4) A and D
- 97.** Unit of evolution is :-
 (1) Species (2) Population
 (3) Individual (4) Community
- 98.** The inner most nutritive wall layer of microsporangium is :-
 (1) Middle layer (2) Exothecium
 (3) Tapetum (4) Endothecium
- 99.** Many insects may consume Pollen and nectar without bringing about pollination. Such floral visitors are referred as :-
 (1) Pollinator (2) Pollen robbers
 (3) Pseudocopulators (4) Translator
- 100.** Filiform apparatus is characteristic feature of :-
 (1) Antipodals (2) Synergids
 (3) Nucellus (4) Endosperm
- 101.** Which of the following structures undergo second meiotic division ?
 (1) Secondary spermatocytes, primary oocyte
 (2) Secondary oocyte, secondary spermatocytes
 (3) Primary spermatocytes, secondary oocyte
 (4) Spermatogonia, Oogonia
- 102.** Which of the following contraceptive methods do involve a role of hormone ?
 (1) Lactational amenorrhoea, pills, emergency contraceptives
 (2) Barrier methods, lactational amenorrhoea, pills
 (3) Cu-T, pills, emergency contraceptives
 (4) Pills, emergency contraceptives, barrier methods
- 103.** In the absence of pregnancy, corpus luteum :-
 (1) degenerates after sometime
 (2) is maintained by progesterone
 (3) becomes active and secretes FSH and LH
 (4) produces oxytocin and relaxin hormone
- 104.** Which of the following statement is correct ?
 (1) After vasectomy, no sperms will be observed in epididymis.
 (2) Failure rate of lactational amenorrhoea is very low.
 (3) Lippe's loop increases the phagocytosis of sperms in uterus
 (4) Male and female condoms inhibits the ovulation and implantation

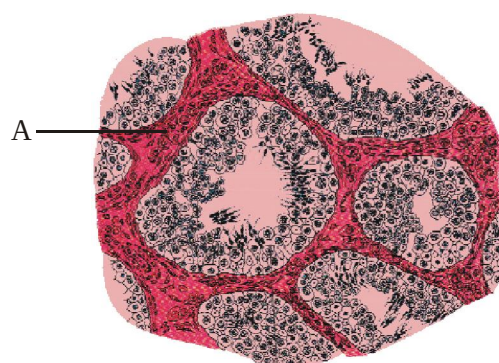
- 105.** Excess use of herbicides and pesticides has only resulted in selection of resistant varieties in a much lesser time scale. It is an example of :-
 (1) Natural selection
 (2) Industrial selection
 (3) Evolution by anthropogenic action
 (4) Both (1) and (3)
- 106.** Which of the following mode of nutrition is most primitive ?
 (1) Chemoautotroph
 (2) Chemoheterotroph
 (3) Aerobic photoautotroph
 (4) Anaerobic photoautotroph
- 107.** Which man fossil is correctly matched with its cranial capacity ?
 (1) *Homo habilis* – 1000 cc
 (2) *Homo erectus* – 900 cc
 (3) Cromagnon man – 1400 cc
 (4) Neanderthal man – 1650 cc
- 108.** The organised typical/polygonum type of embryo sac in angiosperm generally comprises of :-
 (1) Egg, synergids and polar nucleus
 (2) Egg apparatus and polar nucleus
 (3) Egg, synergids and antipodals
 (4) Egg apparatus, antipodals and polar nuclei
- 109.** Development of the embryo directly from the cells of the nucellus is called :-
 (1) Parthenocarpy (2) Parthenogenesis
 (3) Adventive embryony (4) Apogamy
- 110.** Match the column-I and II and choose the correct answer :-

Column-I		Column-II	
a.	Megasporangium	i.	Egg
b.	Megagametophyte	ii.	Embryo-sac
c.	Megasporophyll	iii.	Ovule
d.	Megagamete	iv.	Carpel

Options :-

- (1) a-ii, b-iii, c-i, d-iv (2) a-iv, b-iii, c-ii, d-i
 (3) a-iii, b-ii, c-iv, d-i (4) a-i, b-ii, c-iii, d-iv

- 111.** During menstrual cycle uterine changes are induced by :-
 (1) Pituitary hormones
 (2) Ovarian hormones
 (3) Hypothalamic hormones
 (4) Thyroid hormones
- 112.** Select all the correct to find an answer :-
 (a) MTP is voluntary termination of pregnancy before full term.
 (b) Nearly 45 to 50 billion MTPs are performed in a year all over the world.
 (c) Government of India legalised MTP with some strict conditions in 1971.
 (d) MTP act was amended in year 2017.
 (1) a, b, c are correct, d is wrong
 (2) a, c, d are correct, b is wrong
 (3) a, d are correct, b, c are wrong
 (4) b, d are correct, a, c are wrong
- 113.** Which hormone increases the secretory nature of endometrium ?
 (1) Progesterone (2) Inhibin
 (3) Relaxin (4) Luteinizing hormone
- 114.** In the given figure the function of labelled part-A is :-



- (1) To form male germ cells
 (2) To provide nutrition to germ cells
 (3) To secrete androgens
 (4) To transport sperms

115. Which connotation was not included in the theory of special creation ?

- (1) Earth is about 4000 years old
- (2) All the living organisms passed through gradual changes during evolution
- (3) Diversity was always the same since creation
- (4) All living organisms that we see today were created as such.

116. Flippers of penguins and dolphins are example of :-

- (1) Homologous organ
- (2) Divergent evolution
- (3) Vestigial organ
- (4) Analogous organ

117.

Placental mammal	Marsupial
	
	
	
	
	
	
	

The placental mammals and marsupials of Australia represent which type of evolution ?

- (1) Convergent evolution
- (2) Adaptive radiation
- (3) Divergent evolution
- (4) Both (2) & (3)

118. When pollen grains of a flower is transfer to the stigma of another flower of same plant, the pollination is :-

- (1) Xenogamy
- (2) Geitonogamy
- (3) Homogamy
- (4) Cleistogamy

119. Match the column-I and II and choose correct combination from the given option :-

Column-I		Column-II	
A.	pollen allergy	i.	<i>Citrus</i>
B.	Polyembryony	ii.	Seedless fruit
C.	Parthenocarpy	iii.	<i>Parthenium</i>
D.	False fruit	iv.	Strawberry

- (1) A-iii, B-i, C-ii, D-iv
- (2) A-iii, B-iv, C-i, D-ii
- (3) A-i, B-iv, C-ii, D-iii
- (4) A-i, B-ii, C-iii, D-iv

120. As each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad. So each one is a potential

- (1) Pollen mother cell
- (2) Microspore
- (3) Megaspore mother cell
- (4) Megaspore

121. Choose the correct option regarding 14th day of menstrual cycle :-

- (1) Both LH and FSH attain a peak level.
- (2) FSH, LH and progesterone attain a peak level.
- (3) Both estrogen and progesterone attain a peak level.
- (4) Peak level of FSH & LH and heavy menstruation.

122. In which method of ART, sperm is directly injected into the ovum ?

- (1) IUT
- (2) GIFT
- (3) ICSI
- (4) IUI

123. Which of the following statement is incorrect ?

- (1) Fertilization can only occur if ovum and sperm are transported simultaneously to ampullary region.
- (2) All copulation leads to fertilization and pregnancy.
- (3) Fertilization membrane blocks the entry of additional sperm into ovum.
- (4) Secretion of acrosome helps the sperm to enter into cytoplasm of ovum.

124. Choose the correct option :-

- (1) *Ramapithecus* was more ape like while *Dryopithecus* was more man like
- (2) *Australopithecines* probably lived in East African grasslands.
- (3) *Homo erectus* had a large brain around 1400 cc.
- (4) *Homo erectus* probably did not ate meat.

125. The fitness, according to Darwin, refers ultimately and only to :-

- (1) Adaptive fitness
- (2) Environmental fitness
- (3) Reproductive fitness
- (4) Food and habitat related fitness

126. Ship used by Darwin :-

- (1) HSM beagle
- (2) HMS beagle
- (3) Major ship
- (4) Majesty ship

127. Darwin finche's are example of :-

- (1) Adaptive radiation
- (2) Divergent evolution
- (3) Allopatric speciation
- (4) All of these

128. Double fertilization is found in plants of :-

- (1) Pteridophyta
- (2) Algae
- (3) Angiosperm
- (4) Bryophyta

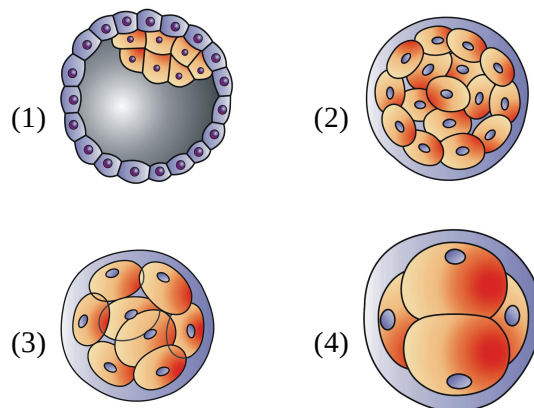
129. Albuminous seeds are found in :-

- (1) Orchid and Castor
- (2) Wheat and maize
- (3) Mango and gram
- (4) Bean and gram

130. The process of development of embryo-sac from a megaspore is known as :-

- (1) Microsporogenesis
- (2) Microgametogenesis
- (3) Megasporogenesis
- (4) Megagametogenesis

131. In the given structure, which one is implanted in endometrium of uterus ?



132. The permissible use of the technique amniocentesis is for :-

- (1) Detecting any genetic abnormality
- (2) Detecting sex of the unborn foetus
- (3) Artificial insemination
- (4) Transfer of embryo into the uterus of a surrogate mother

133. First movment of foetus and appearance of hair on head of foetus is usually observed during which month of pregnancy ?

- (1) Fifth (2) Third (3) Sixth (4) Second

134. Modern giraffes have long neck and long forelimbs because of :-

- (1) internal vital force
- (2) want protection from predators
- (3) nature has selected the giraffes only with long neck and long forelimbs.
- (4) saltation

- 135.** Observe the given table about adaptive radiation of placentals and marsupials; choose the correct option about it :-

Option	Placentals	Marsupials	Type of evolution
(1)	Tasmanian tiger cat	Bobcat	Convergent
(2)	Flying squirrel	Flying phalanger	Divergent
(3)	Lemur	Spotted cuscus	Convergent
(4)	Tasmanian wolf	Tasmanian tiger cat	Covergent

- 136.** The naturalist who came about with the same ideas that of the Darwin :-
 (1) Lamarck (2) De vries
 (3) Alfred Wallace (4) Alfred Nobel
- 137.** Kangaroo and marsupial mole are example of :-
 (1) Divergent evolution
 (2) Adaptive radiation
 (3) Both (1) & (2)
 (4) Convergent evolution
- 138.** In angiosperms normally after fertilization :-
 (1) The zygote divides earlier than the primary endosperm nucleus.
 (2) The primary endosperm nucleus divides earlier than the zygote.
 (3) Both the zygote and primary endosperm nucleus divide simultaneously.
 (4) Both the synergids and primary endosperm nucleus undergo a resting period.
- 139.** After double fertilization embryo sac contain :-
 (1) One diploid and one haploid cell
 (2) Two haploid and one triploid cell
 (3) One diploid and one triploid cell
 (4) One haploid and one triploid cell

- 140.** Pollen grain exine has prominent apertures called:-
 (1) Micropyle (2) Lenticels
 (3) Stomium (4) Germ pores
- 141.** Which of the following group of hormones is secreted by placenta ?
 (1) Estrogens, progestogens, cortisol, hCG
 (2) Estrogens, hCG, hPL, relaxin
 (3) Estrogens, progestogens, thyroxine, hPL
 (4) Estrogens, progestogens, cortisol, thyroxine
- 142.** A contraceptive pill prevents ovulation by :-
 (1) Blocking fallopian tube
 (2) Inhibiting release of FSH & LH
 (3) Stimulating release of FSH & LH
 (4) Causing immediate degeneration of released ovum
- 143.** Which structure helps in transport of substance to and from the embryo ?
 (1) Cervical plug
 (2) Umbilical cord
 (3) Chorionic frondosum
 (4) Chorionic villi
- 144.** Pre-historic cave art was developed about 18,000 years ago. One such cave painting can be seen at :-
 (1) **Bheemlat**, Bundi district
 (2) **Bhimbetka**, Raisen district
 (3) **Beesalpur** dam, Raipur
 (4) **Khajuraho**, Madhya Pradesh
- 145.** The two key concepts of Darwinian theory of evolution are :-
 (1) Use and disuse of organs
 (2) Internal vital force and natural selection
 (3) Branching descent and sexual selection
 (4) Natural selection and branching descent
- 146.** According to Hugo de vries mutations are :-
 (1) Random & directionless
 (2) Small & directionless
 (3) Large & directional
 (4) Random & directional

- 147.** Identify the correct sequence of human evolution ?
 (a) *Australopithecus* (b) *Ramapithecus*
 (c) *Homo erectus* (d) *Homo habilis*
 (e) *Homo sapiens*
 (1) a, b, d, c, e (2) b, a, d, c, e
 (3) a, b, c, e, d (4) b, a, c, e, d
- 148.** Large and often feathery stigma is present in :-
 (1) Entomophilous flowers
 (2) Anemophilous flowers
 (3) Hydrophilous flowers
 (4) Ornithophilous flowers
- 149.** Choose the correct statement from the following:-
 (1) Cleistogamous flowers always exhibit autogamy
 (2) Chasmogamous flowers always exhibit geitonogamy
 (3) Cleistogamous flower exhibit both autogamy and geitonogamy
 (4) Chasmogamous flower never exhibit autogamy
- 150.** The most common type of endosperm in angiosperm is :-
 (1) Free-nuclear endosperm
 (2) Cellular endosperm
 (3) Helobial endosperm
 (4) All of the above
- 151.** Colostrum contain several antibodies but it has abundant :-
 (1) IgA (2) IgD (3) IgG (4) IgM
- 152.** Fructose is secreted from which accessory sex gland?
 (1) Prostate gland (2) Seminal vesicle
 (3) Cowper's gland (4) Bartholin's gland
- 153.** Which of the following statement is incorrect ?
 (1) Seminal plasma is rich in fructose, calcium and certain enzymes.
 (2) Fallopian tube, uterus and vagina constitute female accessory ducts.
 (3) The cleavage division starts as the zygote move through the isthmus of the oviduct towards the uterus
 (4) During embryonic development trophoblast layer gets differentiated as embryo
- 154.** Choose the correct answer about origin of universe :-
 (1) Big-Bang theory attempts to explain origin of life
 (2) The universe is almost 20 million years old.
 (3) Big Bang phenomenon can be explained in physical terms
 (4) Life appeared 500 million years after the formation of earth
- 155.** Human babies birth weight is an example of :-
 (1) Directional selection
 (2) Stabilising selection
 (3) Disruptive selection
 (4) Artificial selection
- 156.** The selection of resistant bacteria and industrial melanism are example of :-
 (1) Natural selection
 (2) Artificial selection
 (3) Mutation
 (4) Adaptive radiation
- 157.** Essential whorls of flowers include :-
 (1) Calyx and epicalyx
 (2) Corolla and calyx
 (3) Androecium and corolla
 (4) Gynoecium and androecium
- 158.** The gametophytic generation in angiosperms is :-
 (1) Short-lived
 (2) Long lived
 (3) Highly reduced
 (4) Both (1) and (3)
- 159.** How many meiosis are required for formation of 100 mature female gametophyte in typical angiosperm ?
 (1) 25 (2) 50 (3) 75 (4) 100
- 160.** In grass embryo, epicotyl has a (a) and a few leaf primordia enclosed in a hollow foliar structure, the (b) .
 (1) (a)-Root apex, (b)-Coleorrhiza
 (2) (a)-Root apex, (b)-Coleoptile
 (3) (a)-Shoot apex, (b)-Coleorrhiza
 (4) (a)-Shoot apex, (b)-Coleoptile

161. Choose the correct statement :-

- (1) FSH act on the sertoli cells and stimulates the secretion of testosterone
- (2) Placenta acts as an endocrine tissue and secretes prolactin
- (3) In the later phase of pregnancy, relaxin is also produced by the ovary
- (4) Immediately after implantation, the outer cell mass (embryo) differentiates into an outer layer called ectoderm

162. How many of the following are not included in external female genitalia ?

- (i) Hymen (ii) Perimetrium
- (iii) Labia majora (iv) Ovary
- (v) Mons pubis (vi) Fundus of uterus
- (vii) Clitoris (viii) Labia majora
- (1) Five (2) Three (3) Two (4) Four

163. Match the column-I and II and choose the correct answer :-

Column-I		Column-II	
A.	Secretory phase	(i)	Formation of corpus luteum
B.	Menstrual phase	(ii)	Formation of mature graafian follicle
C.	Follicular phase	(iii)	Breakdown of endometrium

- (1) A-ii, B-iii, C-i (2) A-i, B-ii, C-iii
- (3) A-i, B-iii, C-ii (4) A-iii, B-i, C-ii

164. According to Oparin and Haldane :-

- (1) life came out of decaying and rotting matter
- (2) life came from outer space
- (3) life came from pre existing living molecules
- (4) life came from pre existing non living organic molecules

165. The first human like being hominid was :-

- (1) *Ramapithecus* (2) *Homo erectus*
- (3) *Australopithecines* (4) *Homo habilis*

166. Who used tools first, in human evolution ?

- (1) Proconsul (2) *Homo habilis*
- (3) *Dryopithecus* (4) *Ramapithecus*

167. A typical anther is :-

- (1) Tetrasporangiate (2) Bisporangiate
- (3) Trisporangiate (4) Monosporangiate

168. A seed is :-

- (1) Unfertilised ovary
- (2) Fertilised ovule
- (3) Unfertilised ovule
- (4) Fertilised ovary

169. Which of the following are correct stages of embryogeny in dicot embryo ?

- (1) Zygote → Globular → Proembryo → Mature embryo
- (2) Proembryo → Globular → Heart shaped → Mature embryo
- (3) Proembryo → Heart shaped → Globular → Mature embryo
- (4) Heart shaped → Mature → Globular → Pro embryo

170. In angiosperms, final product of sexual reproduction is :-

- (1) Ovule (2) Egg
- (3) Zygote (4) Seed

171. Which of the following contraceptive releases Cu-ions and suppress sperm motility ?

- (1) Progestasert (2) LNG-20
- (3) Lippe's loop (4) Multiload 375

172. During oogenesis in mammals oocytes are arrested in which phase of cell division ?

- (1) Interphase (2) Prophase-I
- (3) Metaphase-I (4) Anaphase-II

173. Diaphragms, cervical cap and vaults prevents conception by :-

- (1) Suppressing the release of FSH & LH
- (2) Increasing phagocytosis of sperm within uterus
- (3) Blocking entry of sperm through cervix
- (4) Inhibiting ovulation

174. Which is not a pair of analogous structures ?

- (1) Eye of the octopus and mammals
- (2) Flippers of penguins and dolphins
- (3) Wings of butterfly and birds
- (4) Forelimbs of whale, bats, cheetah and human

175. Observe the given table and choose the correct match:-

Column-I		Column-II	
(A)	<i>Homo habilis</i>	(i)	Probably did not ate meat
(B)	Java man	(ii)	Used animal hide to protect body
(C)	Neanderthal man	(iii)	Pre historic cave art
(D)	Cromagnon man	(iv)	First use of fire

- (1) A-(i), B-(ii), C-(iv), D-(iii)
- (2) A-(i), B-(iv), C-(ii), D-(iii)
- (3) A-(ii), B-(i), C-(iv), D-(iii)
- (4) A-(ii), B-(iv), C-(iii), D-(i)

176. Match the following columns :-

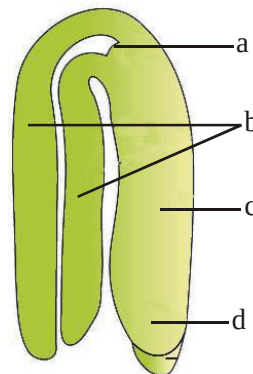
Column-I		Column-II	
(A)	Darwin	(i)	Abiogenesis
(B)	Oparin	(ii)	Use or disuse of organs
(C)	Lamarck	(iii)	Continental drift theory
(D)	Wegner	(iv)	Natural selection

- (1) A-(i), B-(iv), C-(ii), D-(iii)
- (2) A-(iv), B-(i), C-(ii), D-(iii)
- (3) A-(ii), B-(iv), C-(iii), D-(i)
- (4) A-(iv), B-(iii), C-(ii), D-(i)

177. Male gametophyte of angiosperms/monocots is :-

- (1) Microsporangium
- (2) Nucellus
- (3) Pollen grains
- (4) Stamen

178. Recognise the figure and find out the correct matching:-



- (1) a-Radicle, b-Hypocotyl, c-Epicotyl, d-Plumule
- (2) a-Plumule, b-Cotyledon, c-Hypocotyl, d-Radicle
- (3) a-Epicotyl, b-Cotyledon, c-Plumule, d-Radicle
- (4) a-Epicotyl, b-Cotyledon, c-Radicle, d-Plumule

179. How many of these are examples of non-endospermic seeds ?

Gram, Pea, Bean, Groundnut, Castor, Wheat, Rice, Coconut

- (1) Four
- (2) Five
- (3) Six
- (4) Seven

180. Perisperm is found in the :-

- (1) Root of beet
- (2) Seed of beet
- (3) Embryo of beet
- (4) Stem of beet

SPACE FOR ROUGH WORK

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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